
Exercise sheet 7 for Numerical Optimization

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Exercise 1 (Convergence statement (Zoutendijk) of descent methods, 9+9+9 points)

You already got a convergence statement of descent methods in the lecture. Now you will get to know a second one, the theorem of Zoutendijk. It demands more assumptions and yields classical convergence to a stationary point, not only that every accumulation point is a stationary point.

It states the following: Let $x_0 \in \mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be bounded from below. Let $(x_k)_{k \in \mathbb{N}_0} \subset \mathbb{R}^n$ be the iterates of a descent method and $\Omega \subset \mathbb{R}^n$ an open set in such a way that $(x_k)_{k \in \mathbb{N}_0} \subset \Omega$. Assume that f is continuously differentiable on Ω and that ∇f is Lipschitz continuous on Ω with Lipschitz constant $L > 0$. If

- the step sizes $\alpha_k > 0$ of all iteration steps fulfil the Wolfe condition,
- the directions $d_k \in \mathbb{R}^n$ of all iteration steps are search directions, so $\nabla f(x_k)^T d_k < 0$,
- and the angles θ_k between the search directions d_k and the gradients $\nabla f(x_k)$ are uniformly bounded away from 90° , meaning that there exists a $\delta > 0$ such that

$$\cos(\theta_k) = \frac{-\nabla f(x_k)^T d_k}{\|\nabla f(x_k)\| \|d_k\|} \geq \delta, \quad (1)$$

it holds true that

$$\lim_{k \rightarrow \infty} \|\nabla f(x_k)\| = 0.$$

- a) Prove that there is $C > 0$ such that

$$f(x_{k+1}) \leq f(x_k) - C \cos(\theta_k)^2 \|\nabla f(x_k)\|_2^2.$$

- b) Prove that

$$\sum_{k=0}^{\infty} \cos(\theta_k)^2 \|\nabla f(x_k)\|_2^2 < \infty.$$

- c) Use b) to prove the Zoutendijk convergence statement.

Exercise 2 (Zoom algorithm for strong Wolfe conditions, 9+9 points)

The Zoom algorithm is a common algorithm for determining a step size α that satisfies the strong Wolfe conditions for an iterate x and a search direction d . Remember that on sheet 5 exercise 3b you implemented an algorithm for finding a step size that satisfies the Armijo condition (the so-called *backtracking algorithm*). The Zoom algorithm is:

Algorithm 1: Zoom algorithm for strong Wolfe conditions

```
1 Input:  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ ,  $x \in \mathbb{R}^n$ ,  $d \in \mathbb{R}^n$ ,  $\alpha_{\text{init}} > 0$ ,  $0 < c_1 < c_2 < 1$ 
2 Output: step size  $\alpha > 0$  satisfying the strong Wolfe conditions
3  $\alpha_0 \leftarrow 0$ 
4  $\alpha_1 \leftarrow \alpha_{\text{init}}$ 
5  $i \leftarrow 1$ 
6 while True do
7   if  $f(x + \alpha_i d) > f(x) + c_1 \alpha_i \nabla f(x)^T d \vee f(x + \alpha_i d) \geq f(x + \alpha_{i-1} d)$  then
8     return zoom( $\alpha_{i-1}$ ,  $\alpha_i$ )
9   if  $|\nabla f(x + \alpha_i d)^T d| \leq c_2 |\nabla f(x)^T d|$  then
10    return  $\alpha_i$ 
11   if  $\nabla f(x + \alpha_i d)^T d \geq 0$  then
12    return zoom( $\alpha_i$ ,  $\alpha_{i-1}$ )
13    $\alpha_{i+1} \leftarrow 2\alpha_i$ 
14    $i \leftarrow i + 1$ 
15
16 function  $\alpha = \text{zoom}(\beta_1, \beta_2)$ 
17 while True do
18    $\alpha \leftarrow \frac{\beta_1 + \beta_2}{2}$ 
19   if  $f(x + \alpha d) > f(x) + c_1 \alpha \nabla f(x)^T d \vee f(x + \alpha d) \geq f(x + \beta_1 d)$  then
20      $\beta_2 \leftarrow \alpha$ 
21   else
22     if  $|\nabla f(x + \alpha d)^T d| \leq c_2 |\nabla f(x)^T d|$  then
23       return  $\alpha$ 
24     if  $\nabla f(x + \alpha d)^T d (\beta_2 - \beta_1) \geq 0$  then
25        $\beta_2 \leftarrow \beta_1$ 
26      $\beta_1 \leftarrow \alpha$ 
```

- a) Understand what the algorithm is doing. Since the algorithm is rather tricky, it is sufficient to understand its essentials.

Hint: 1) When calling `zoom`(β_1, β_2), it is ensured that there is an interval with step sizes satisfying the strong Wolfe conditions between β_1 and β_2 . Make yourself clear why.

2) Make yourself clear that the inner function `zoom` is performing a bisection.

- b) Create a MATLAB programme `zoom_wolfe.m` that implements algorithm 1. It should be passed a function handle `f`, a function handle `grad_f`, an iterate $x \in \mathbb{R}^n$, a search direction $d \in \mathbb{R}^n$ and parameters $\alpha_{\text{init}} > 0$ and $0 < c_1 < c_2 < 1$.

The programme will be tested in the next exercise.

Exercise 3 (Comparison of descent methods and step size strategies using Rosenbrock's function, 5 points)

As grand finale to the descent methods, we will now compare all descent method variants we have looked at - gradient descent, Newton, nonlinear cg (Fletcher-Reeves) - using both step size strategies we have implemented, backtracking for an Armijo step size and the zoom algorithm for a strong Wolfe step size, using Rosenbrock's function.

Therefore, execute the MATLAB programme `comparison_descent_methods.m` and reflect on the results.